

Dynamical Analysis of a Nonlinear State Variable Filter

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A nonlinear 3rd order state variable filter (SVF) is displayed in figure 1. The system was adapted from a simple nonlinear oscillator proposed by J.C. Sprott. [1] The nonlinearity was chosen because it is odd and non-monotonic. The odd nature of this function preserves bipolarity, which is necessary for audio signals. Non-monotonicity introduces a folding mechanism into the system's phase-space dynamics, allowing trajectories to be stretched and folded in a manner characteristic of chaotic systems. A simple example of this mechanism can be seen in the logistic map. [2]

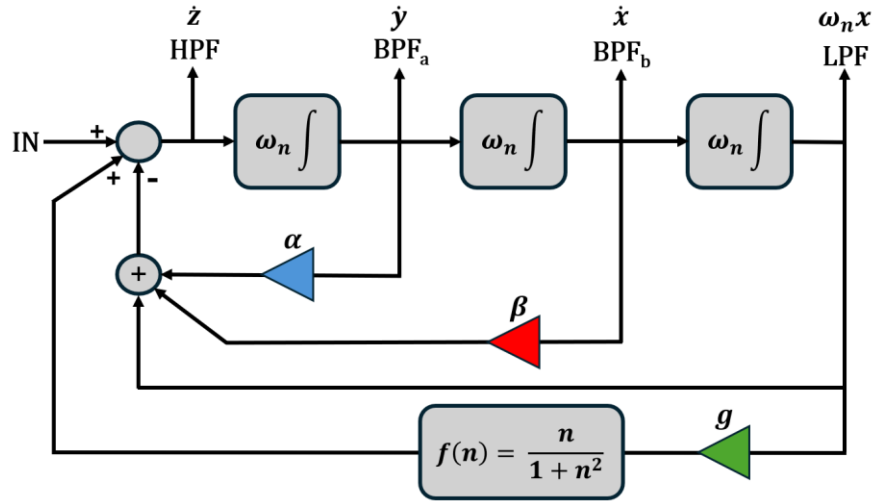


Fig. 1: Nonlinear 3rd Order SVF.

$$\begin{aligned} \dot{z} &= In - \alpha \omega_n z - \beta \omega_n y - \left(1 - \frac{g}{1 + (\omega_n x)^2}\right) \omega_n x \\ \dot{y} &= \omega_n z \\ \dot{x} &= \omega_n y \end{aligned} \tag{1}$$

The Jacobian of this system may be expressed as

$$J = \begin{bmatrix} 0 & \omega_n & 0 \\ 0 & 0 & \omega_n \\ \omega_n \left(-1 + \frac{g(1 - (\omega_n x)^2)}{(1 + (\omega_n x)^2)^2} \right) & -\beta \omega_n & -\alpha \omega_n \end{bmatrix}.$$

Setting the input to zero, the fixed points of the system exist at $(x, y, z) = (0, 0, 0)$ and $(\pm \frac{1}{\omega_n} \sqrt{g-1}, 0, 0)$. Fixed point analysis may be performed to understand the bifurcations that characterize the nonlinear filter.

At the origin, the Jacobian may be simplified to

$$J(0,0,0) = \begin{bmatrix} 0 & \omega_n & 0 \\ 0 & 0 & \omega_n \\ \omega_n(g-1) & -\beta\omega_n & -\alpha\omega_n \end{bmatrix}.$$

The resulting characteristic equation is

$$\lambda^3 + \lambda^2\alpha\omega_n + \lambda\beta\omega_n^2 + \omega_n^3(1-g) = 0.$$

From the Routh-Hurwitz stability criterion, cubic polynomials of the form $p(x) = x^3 + ax^2 + bx + c$ must satisfy the following conditions for the eigenvalues to have negative real parts. [3]

1. $a, b, c > 0$
2. $ab > c$

It can be shown that if condition 1 is satisfied, but condition 2 is not, then the purely real eigenvalue is negative, but the complex conjugate eigenvalues have $\Re\{\lambda\} > 0$.

Applying these criteria to the characteristic equation of $J(0,0,0)$, the system is stable at the origin when $\alpha, \beta > 0$, $g < 1$, and $\alpha\beta > 1 - g$. For the underlying linear system ($g = 0$), this condition simplifies to $\alpha\beta > 1$. For this reason, the α and β parameters were chosen to be greater than or equal to 1 in the implemented filter. At $\alpha, \beta = 1$ and $g = 0$, the complex poles reach the imaginary axis, and the filter self oscillates. The origin becomes unstable when g is increased to 1.

At the $(\pm \frac{1}{\omega_n} \sqrt{g-1}, 0, 0)$ fixed point, the Jacobian simplifies to

$$J\left(\pm \frac{1}{\omega_n} \sqrt{g-1}, 0, 0\right) = \begin{bmatrix} 0 & \omega_n & 0 \\ 0 & 0 & \omega_n \\ 2\omega_n\left(\frac{1}{g}-1\right) & -\beta\omega_n & -\alpha\omega_n \end{bmatrix},$$

And the characteristic equation of $J\left(\pm \frac{1}{\omega_n} \sqrt{g-1}, 0, 0\right)$ is

$$\lambda^3 + \lambda^2\alpha\omega_n + \lambda\beta\omega_n^2 + 2\omega_n^3\left(1 - \frac{1}{g}\right) = 0.$$

Applying the Routh-Hurwitz stability criteria, the $\left(\pm \frac{1}{\omega_n} \sqrt{g-1}, 0, 0\right)$ fixed points require $1 < g < \frac{2}{2-\alpha\beta}$. These results support the existence of a supercritical pitchfork bifurcation at $g = 1$. When g is increased to 1, the stability of the origin is transferred to the $\pm \frac{1}{\omega_n} \sqrt{g-1}$ branches, which serve as the equilibrium points around which the attractors develop. The $g < \frac{2}{2-\alpha\beta}$ condition is imposed by the second Routh-Hurwitz criterion. At this point, the complex conjugate eigenvalues cross the imaginary axis, resulting in a Hopf bifurcation. At g values beyond this threshold, a double-scroll attractor emerges. This bifurcation sequence is similar to that of the Lorenz system, though the underlying dynamics differ considerably.

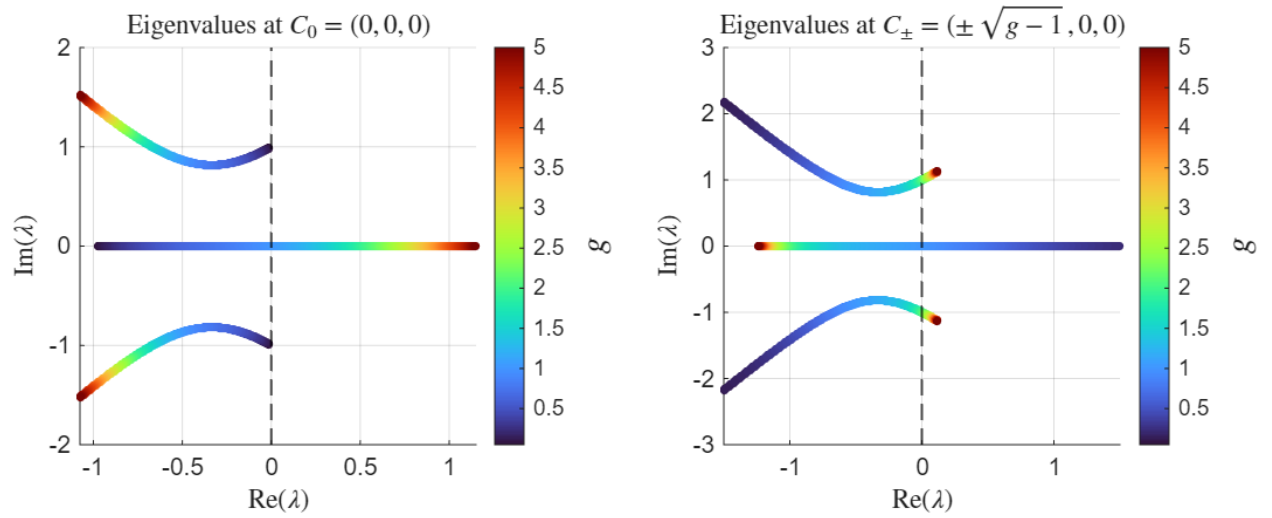


Fig. 2: Root Locus plot displaying eigenvalues as a function of the nonlinearity gain with $\alpha, \beta = 1$.

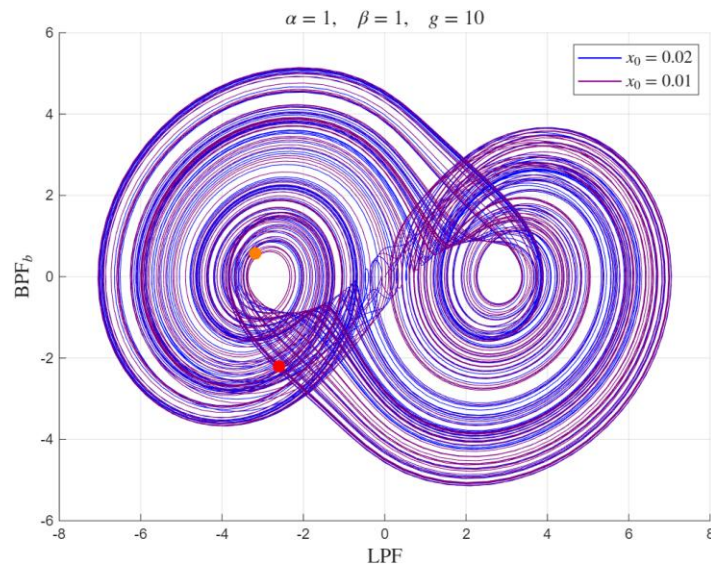


Fig. 3: Double-scroll attractor.

References:

1. W. M. Ahmad and J. C. Sprott, "Chaos in fractional-order autonomous nonlinear systems," *Chaos, Solitons & Fractals*, vol. 16, no. 2, pp. 339–351, Mar. 2003, doi: 10.1016/S0960-0779(02)00438-1.
2. E. W. Weisstein, "Logistic Map," MathWorld—A Wolfram Resource. [Online]. Available: <https://mathworld.wolfram.com/LogisticMap.html>. [Accessed: Aug. 19, 2026].
3. M. Raginsky and D. Liberzon, "ECE 486 Control Systems Lecture 7," ECE 486 Handbook, University of Illinois Urbana-Champaign, 2018.